MCS-212	WIRELESS SENSOR NETWORKS	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To study the concepts of sensor networks.
CO2	To understand the WSN node Architecture and Newtork Architecture
CO3	To study the research issues in different layers of sensor networks
CO4	To identify the Wireless Sensor Network Platforms
CO5	To design and Develop wireless sensor node

MODULE I

Introduction and overview: Overview of the course; overview of sensor network protocols, architecture, and applications; simulation and experimental platforms; main features of WSNs; research issues and trends.

Enabling technologies : Fundamentals of 802.15.4, Bluetooth, and UWB; Physical and MAC layers.

MODULE II

Sensor node hardware and software : Hardware: mica2, micaZ, telosB, cricket, Imote2, tmote, btnode, and Sun SPOT. Software (OS): tinyOS, MANTIS, Contiki, and RetOS. Programming tools: C, nesC, Mate, Localization, connectivity, and topology, Sensor deployment mechanisms; coverage issues; node discovery protocols.

MODULE III

Network layer protocols : Data dissemination and processing; multi-hop and cluster based protocols; routing.

Middleware and application layers, Data dissemination; data storage; query processing; sensorWeb; sensorGrid.

Open issues for future research, Energy preservation and efficiency; security challenges; fault-tolerance;

References

1. *H. Karl and A. Willig. "Protocols and Architectures for Wireless Sensor Networks", John Wiley & Sons, June 2005.*
2. *K. Sohraby, D. Minoli, and T. Znati, "Wireless Sensor Networks: Technology, Protocols, and Applications", John Wiley & Sons, March 2007.*
3. *C. S. Raghavendra, K. M. Sivalingam, and T. Znati, "Wireless Sensor Networks", Springer Verlag, Sep. 2006.*
4. *E. H. Callaway, "Wireless Sensor Networks: Architectures and Protocols", Jr. AUERBACH, Aug. 2003.*

5. *B. Krishnamachari "Networking Wireless Sensors", Cambridge University Press, Dec. 2005.*
6. *F. Zhao and L. Guibas "Wireless Sensor Networks: An Information Processing Approach", Morgan Kaufmann, Jul. 2004.*
7. *N. P. Mahalik, "Sensor Networks and Configuration: Fundamentals, Standards, Platforms, and Applications", Springer Verlag, Nov. 2006.*
8. *N. Bulusu and S. Jha, "Wireless Sensor Networks: A Systems Perspective", Artech House, August 2005*

Course Outcome:

CO1	Explain the basic concepts of wireless sensor networks, sensing, computing and communication tasks
CO2	Describe and explain radio standards and communication protocols adopted in wireless sensor networks
CO3	Describe and explain the hardware, software and communication for wireless sensor network nodes
CO4	Explain the architectures, features, and performance for wireless sensor network systems and platforms
CO5	Describe and analyze the specific requirements of applications in wireless sensor networks for energy efficiency, computing, storage and transmission